

SOFTWARE REQUIREMENT SPECIFICATION

FOR

AIRLINE MANAGEMENT SYSTEM

Under the Subject of

Software Engineering

(Semester - I)

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CERTIFICATE

Date:

This is to certify that the Software Engineering Work entitled “**Airline Management System**”, carried out by the group of students mentioned below under my guidance is approved for the Degree of Bachelor of Engineering (Semester-I) of L.J. Institute of Engineering and Technology (LJU) during the academic year 2024-25.

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Guide

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1. INTRODUCTION

1.1 Purpose:

An Airline Management System (AMS) is a fully integrated software package that manages airlines' different functions, including system login, flight scheduling , reservations, ticketing, passenger and crew management, cargo handling, and aircraft maintenance. It enhances the operational efficiency, improves the customer experience, ensures regulatory compliance, and maximizes resource utilization for airlines.

1.2 Scope:

The airline booking website is an application stored in the user server. The purpose of the website is to resolve the client to allow website users to perform tasks related to visit website, booking tickets and canceling tickets of an airline flight. The system enables to perform the following functions:

- Login/Sign in
- Automation of flight operations
- Automation of ticketing / seat booking
- confirmation system
- Cancellation
- Improved and optimized service

1.3 Glossary:

- AMS-Airline Management System
- LAN-Local Area Network
- GUI-Graphical User Interface
- OS-Operating System
- RAM-Random Access Memory
- MB-Mega Bytes
- GB-Giga Bytes
- Mbps-Megabits per second
- HDD-Hard Disk Drive 1
- UML-unified modelling language

1.4 Overview:

The remaining section of this document provide a general description including characteristic of the users of this product, the product's hardware, and functional and non-functional requirements of the product.

2. OVERALL DESCRIPTION

2.1 Problem Statement

Developing an AIRLINE MANAGEMENT SYSTEM :- AMS for an airline company that want to automate its Login/Sign features and automate in flight operations and ticketing / seat booking and confirmation system and Cancellation.

2.2 Existing System

Before the automation the system suffered from following DRAWBACKS:

- Existing system is highly manual and involves a lot of paper work and calculation and
- therefore, may be erroneous. This led to inconsistency and inaccuracy.
- The data may be lost, stolen or destroyed because it is stored on paper.
- The existing system consumes a lot of time causing inconveniencing to customers and the staff.
- It's difficult to update, delete, or view the data due its manual nature.
- Increasing number of passengers leads to difficulty in maintaining and retrieving details.
- It is time consuming.

2.3 Proposed System

The AMS is proposed with the following:

- The computerization of the management system will reduce a lot of paperwork and hence the load on airline admin and staff.
- The machine will perform all calculations. Hence chances of error are nearer to null.
- The passenger Login/Sign in, reservation, cancellation list can be easily retrieved and any required
- addition, deletion, updation can be performed easily and fast.
- Proper way of confirmation of bookings etc.

2.4 Product Functions

Booking agents with varying levels of familiarity with computers will mostly use this system. With this on mind, an important feature of this software is that it will be relatively simple to use. The scope of this product encompasses:

SEARCH: This function allows the booking agent to search for aeroplane's and ticket's availability between two cities, i.e. departure city and arrival city, the date of departure, preferred time and number of passengers.

2.5 User characteristics

(2.5.1) User Role

- Passenger : Family traveler , frequent flyer , international traveler , business traveler , casual traveler.
- Travel Agent : Individual agent , cooperate travel agent , online booking platform.
- Airline Staff : Reservation agent , check in counter staff , gate agent , customer service representative , flight crew.
- Admin Staff : System administrator , data analyst.

(2.5.2) Technical Expertise

- Basic : User comfortable with basic web browsing and online forms.
- Intermediate : Users familiar with online booking platforms and may require more advanced search options.
- Advanced : Users with experience in travel management systems, potentially needing access to complex reporting features.

(2.5.3) Usage Frequency

- Occasional: Users who book flights infrequently.
- Regular: Users who book flights frequently for business or personal travel.

(2.5.4) User's Accessibility Needs

- Visual impairments
- Hearing impairments
- Motor disabilities

2.6 Constraints

• Performance constraints:

- Real-time flight status updates must be available within a specified timeframe.
- The system must handle a high volume of concurrent user requests during peak booking periods.
- Response time for critical operations like ticket cancellation should be minimal.

• Data constraints:

- Passenger data storage limitations due to privacy regulations.
- Database size restrictions for flight schedules and inventory management.
- Data integrity rules for sensitive passenger information.

• Integration constraints:

- Compatibility with existing reservation systems used by travel agencies.

- Interfacing with third-party payment gateways for secure transactions.
- Integration with airport management systems for check-in and boarding processes.
- **Security constraints:**
 - Compliance with industry standards like PCI DSS for payment data protection.
 - Implementation of strong user authentication mechanisms to prevent unauthorized access.
 - Encryption of sensitive passenger data at rest and in transit.
- **Regulatory constraints:**
 - Adherence to FAA regulations regarding flight schedules and passenger information.
 - Compliance with local data privacy laws in different operating regions.
- **Budgetary constraints:**
 - Limitations on development costs for specific features.
 - Restrictions on hardware and software purchases due to budget limitations.
- **Time constraints:**
 - Project deadlines for system launch and deployment.
 - Time allocated for testing and quality assurance phases.

2.7 Assumptions and Dependencies

➤ Assumptions:

- The airline's existing management system will provide accurate and timely data for integration with the new management system.
- The system will be implemented on a secure server with appropriate data encryption measures.
- Flight schedules and bookings assume proper handling of different time zones.

➤ Dependencies:

- The system should run smoothly on Windows, macOS, and Linux.
- External authentication services (OAuth, Google Sign-in) are used for user login.
- The system depends on adequate server infrastructure to handle requests efficiently.
- External backup solutions are required to prevent data loss.
- The system assumes airlines will provide accurate flight and pricing details for bookings.

3. REQUIREMENT SPECIFICATION

3.1 Functional Requirements

1. User Management

- Users can register and log in (Passengers, Administrators, Travel Agents).
- Role-based access control (Passenger, Admin, Agent).
- Password recovery and profile management.

2. Flight Search & Booking

- Search flights based on origin, destination, date, and number of passengers.
- Filter results by airline, class (Economy, Business, First), price, and layovers.
- Select available seats from the seating plan.
- Add baggage options during booking.
- Provide fare details, taxes, and additional charges before confirmation.

3. Payment Processing

- Multiple payment methods (Credit/Debit Card, PayPal, Wallet, UPI, Net Banking).
- Secure payment gateway integration.
- Generate and send invoices/receipts via email.
- Currency conversion for international bookings.

4. Ticket Management

- Generate electronic tickets (E-Tickets).
- View, download, and print tickets.
- Email confirmation and SMS notifications.
- QR code generation for mobile boarding passes.

5. Booking Modification & Cancellation

- Allow customers to modify travel dates, seat selection, and add-ons.
- Cancellation policies with refund processing.

- Partial or full refunds based on the airline's rules.

3.1.1 Performance Requirement

1. System Response Time

- Flight Search: Results should be displayed within 2-3 seconds.
- Booking Confirmation: Transaction processing should take less than 5 seconds.
- Payment Processing: Secure payment transactions should complete within 7 seconds.
- Check-in Processing: Boarding passes should be generated within 3 seconds.
- Admin Reports: Large data reports should generate within 10 seconds.

2. Scalability

- Must handle millions of users concurrently during peak times (e.g., holiday seasons).
- Support for horizontal scaling (adding more servers) to accommodate increasing demand.
- Load balancing across multiple servers to prevent bottlenecks.

3. Availability & Reliability

- 99.99% uptime guarantee, ensuring continuous operation with minimal downtime.
- Automatic failover mechanisms in case of server failure.
- Database replication for redundancy to avoid data loss.

3.1.2 Design Requirement

- An architecture built on microservices for modularity and scalability.
- Load balancing to effectively manage high traffic volumes.
- Horizontal scaling to handle growing traffic volumes.

3.1.3 Software Requirement

- Operating System-Windows 11
- Notepad++ - For coding
- Word
- Power point
- Lucid Charts – For Diagrams

3.1.4 Other Requirement

- Multi-Language Support
- Accessibility Compliance
- User-Friendly Design:
 - Steps for booking and check-in.
 - Mobile-friendly UI with fast loading times.
- Data Backup Strategy:
 - Real time repetition + daily full backups

3.2 Non-Functional Requirements

- Paperless Operations: Digital tickets, e-boarding passes, and e-receipts.
- Energy-Efficient Cloud Computing to reduce carbon footprint.
- System Response Time:
 - Search Flights: Results should be displayed within 2–3 seconds.
 - Booking Process: Must complete within 5 seconds.
 - Payment Processing: Should not exceed 7 seconds.
 - Check-in Processing: Boarding pass generation within 3 seconds.
- Large reports should generate within 10 seconds.
- Must support 100,000+ concurrent users without performance degradation.
- System should handle 1,000+ transactions per second during peak hours.
- Data Synchronization: Real-time updates across different airline systems.
- ACID Compliance: Transactions should be atomic to prevent double booking.

3.2.1 Security

- Authentication & Authorization
- Data Security & Encryption
- Network & Infrastructure Security
- Payment Security
- Threat Detection & Incident Response
- Compliance & Regulatory Requirements:
- Airline Industry Regulations

3.2.2 Reliability

- High Availability (HA) & Uptime
- Data Integrity & Consistency
- Performance & Scalability

3.2.3 Availability

- High Availability (HA) Architecture
- Continuous Monitoring & Predictive Maintenance

3.2.4 Maintainability

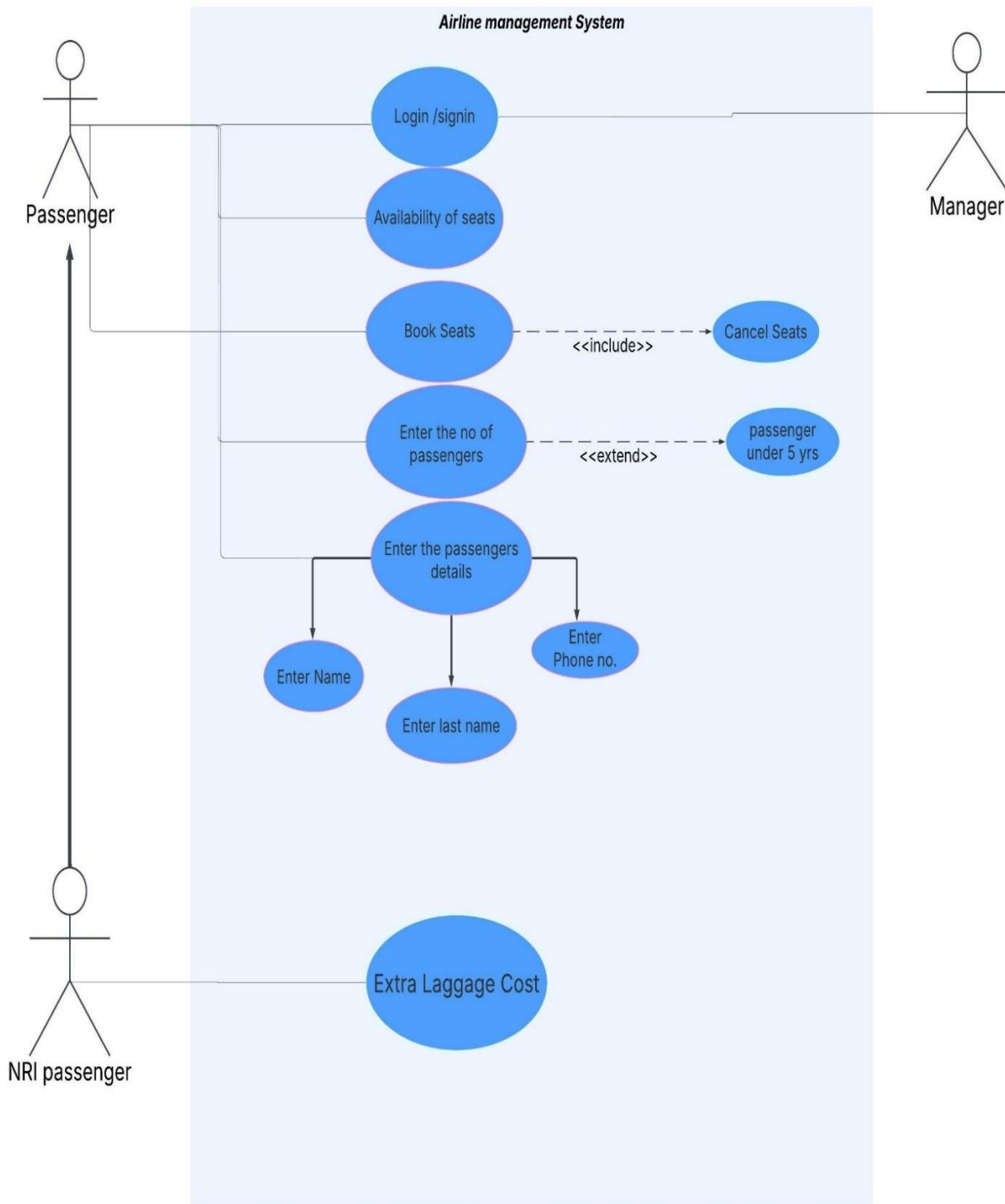
- Code Maintainability & Best Practices
- Automated Monitoring & Logging
- Database Maintainability
- Security & Compliance Maintenance

3.2.5 Supportability

- System Monitoring & Alerts
- Technical Support & Help Desk
- Software Maintainability & Updates

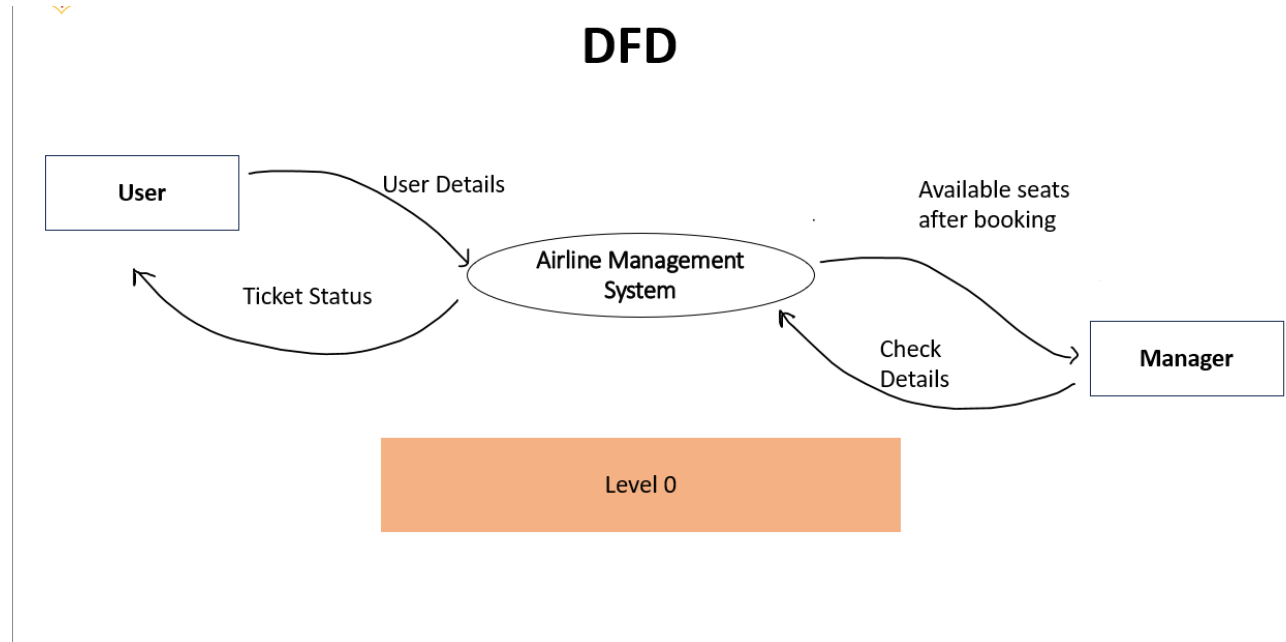
4. DIAGRAMS

4.1 Use Case Diagram

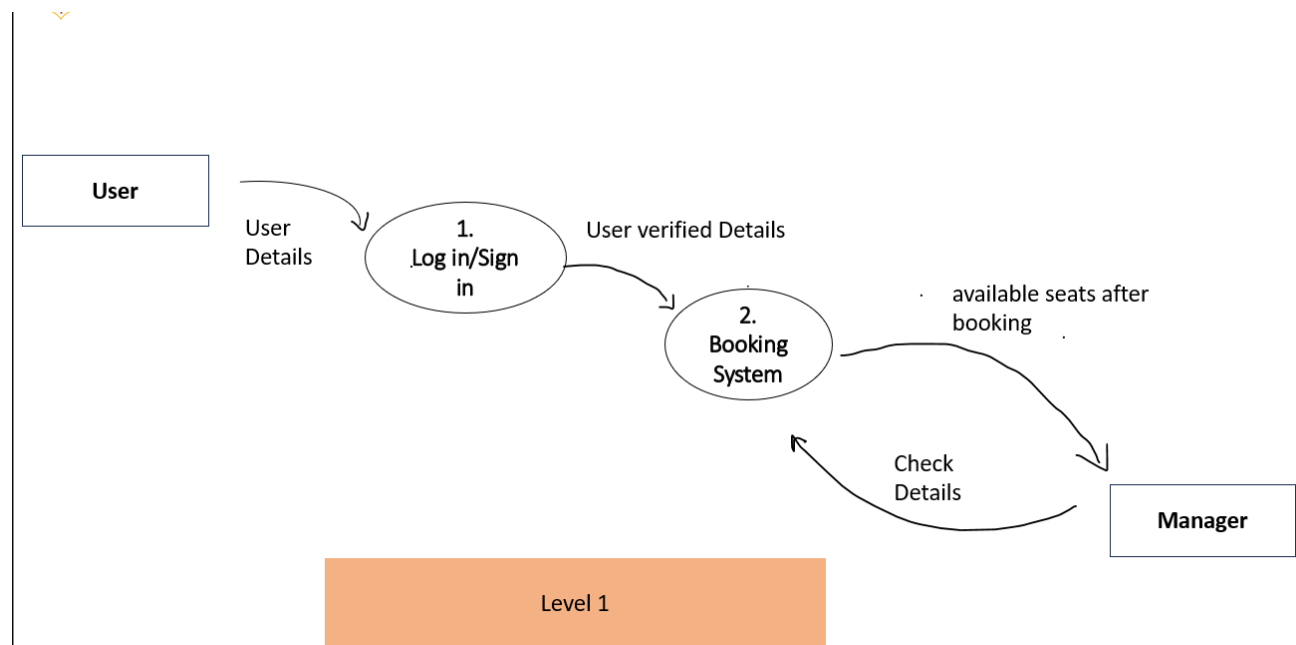


4.2 Data Flow Diagram

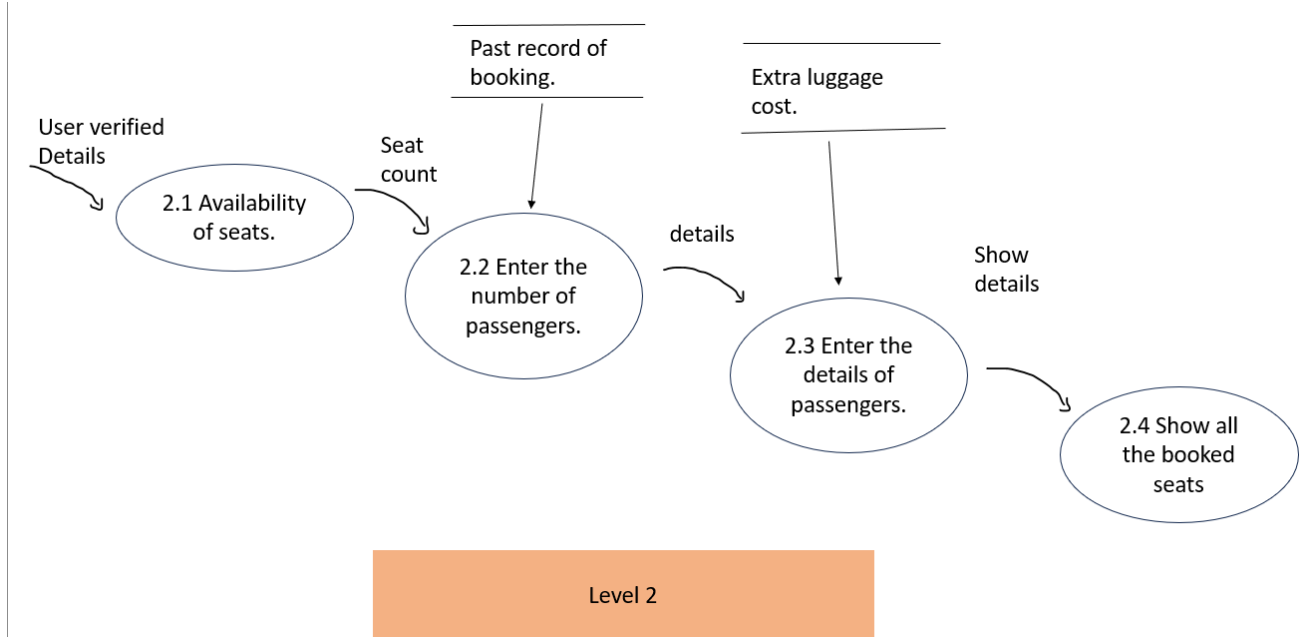
4.2.1 Level-0 DFD



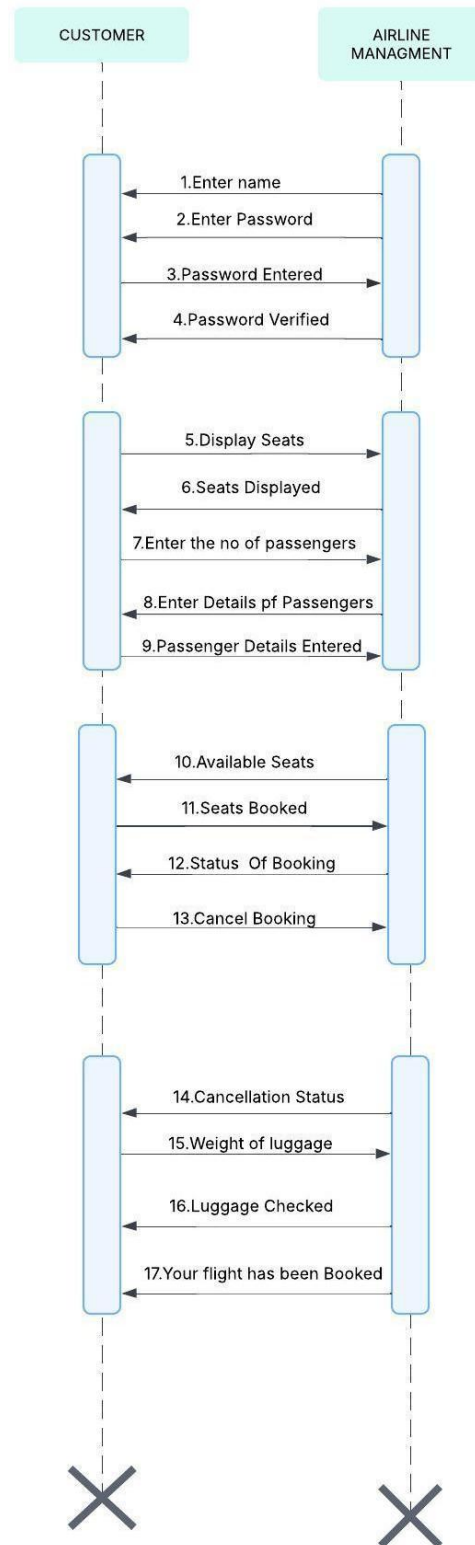
4.2.2 Level-1 DFD



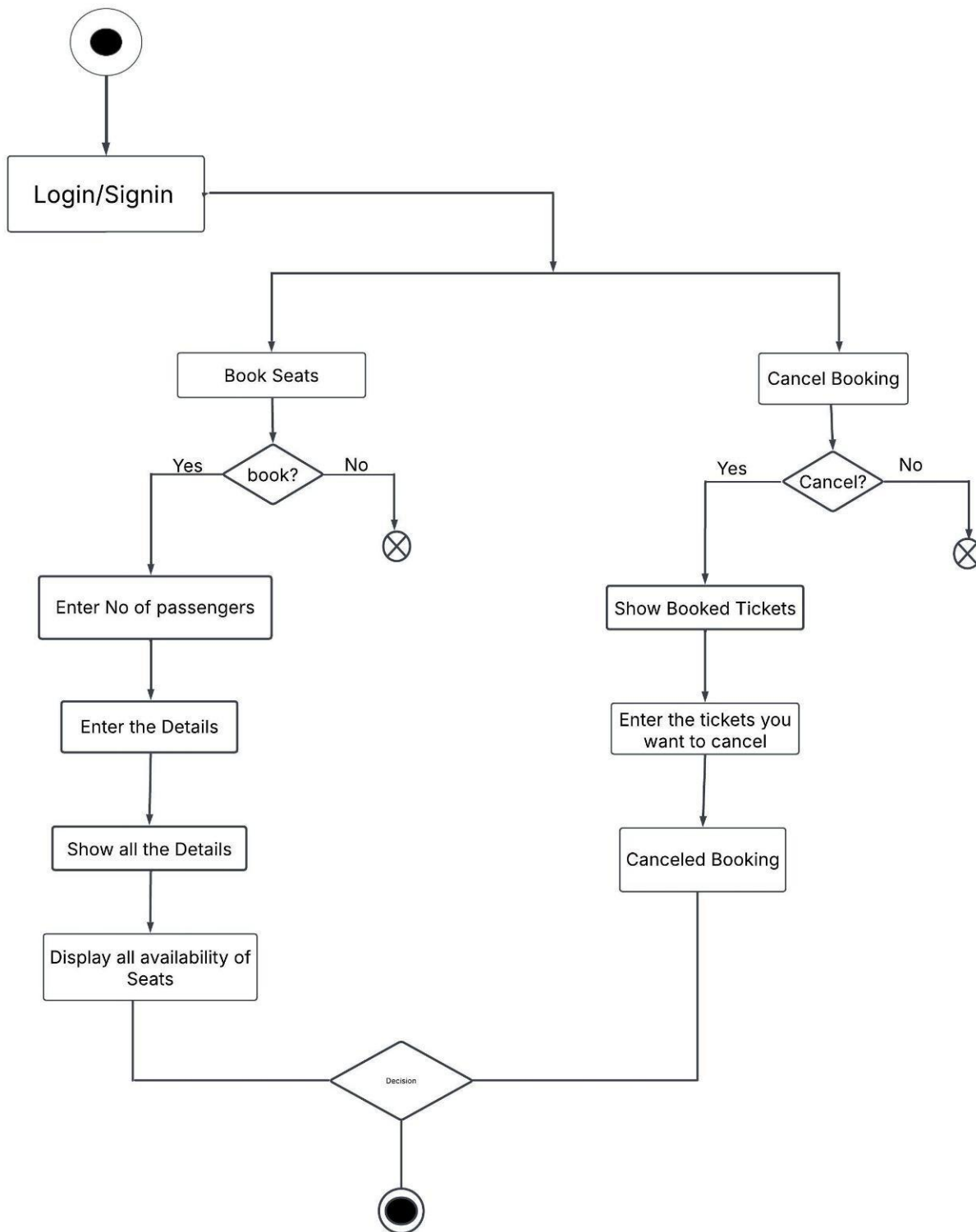
4.2.3 Level-2 DFD



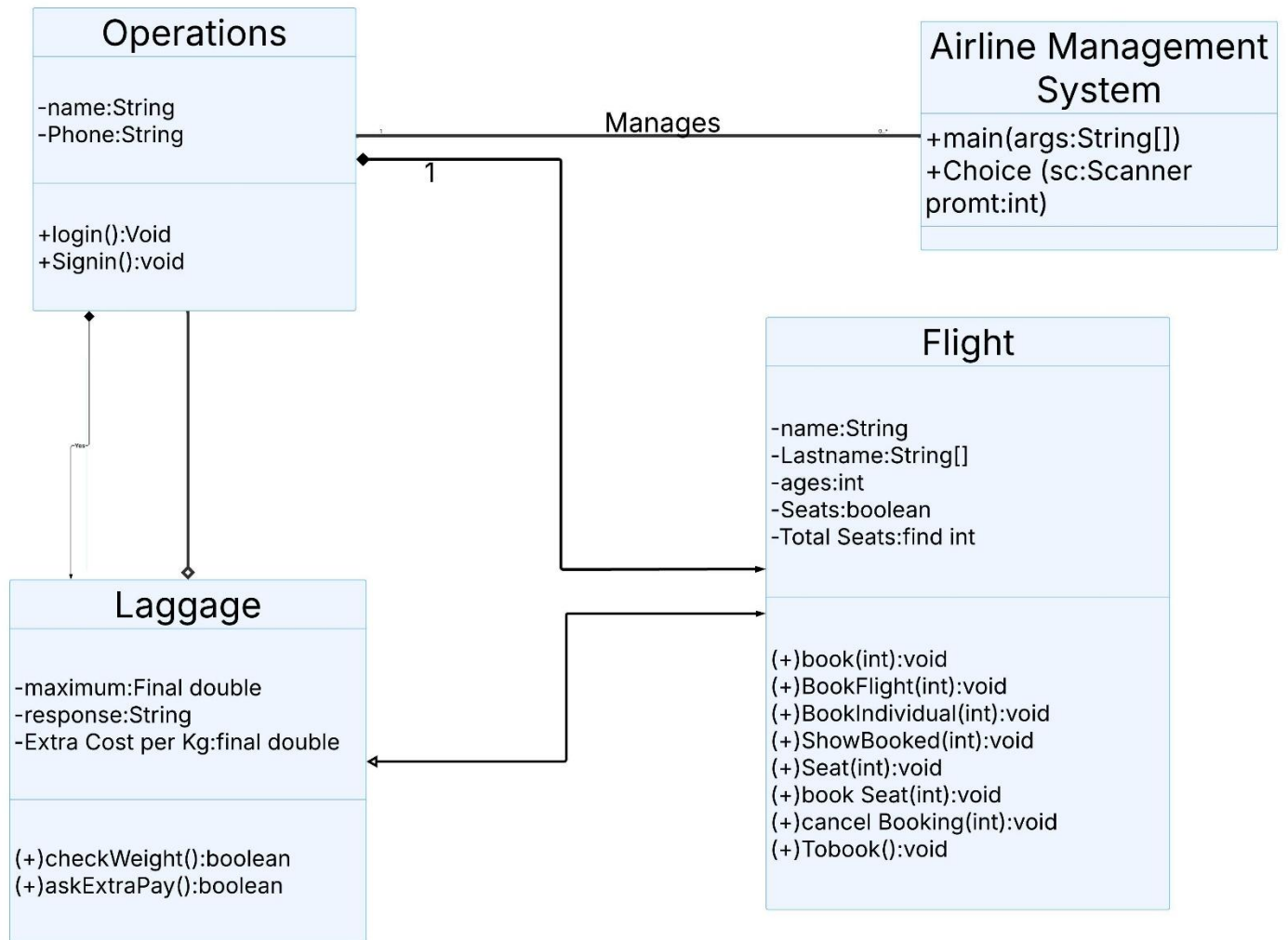
4.3 Sequential Diagram



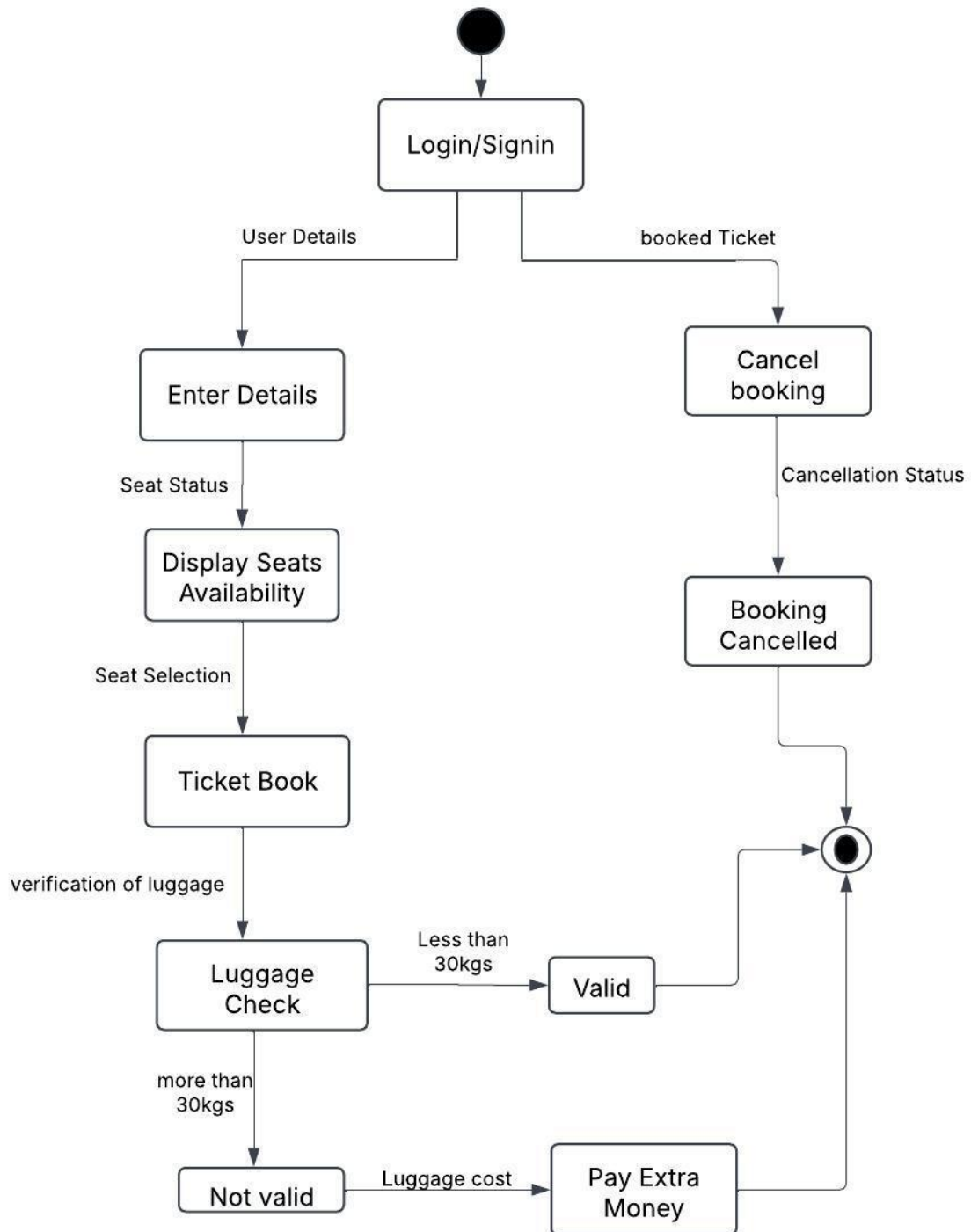
4.4 Activity Diagram



4.5 Class Diagram



4.6 State Diagram



5. CONCLUSION

An effective Airline Management System is crucial for the competitiveness and sustainability of airlines in a fast-paced, dynamic industry. By automating and optimizing core functions, AMS can enhance efficiency, improve customer service, and ultimately contribute to an airline's profitability. However, airlines must carefully select, customize, and maintain their AMS to ensure it aligns with their specific operational needs and can adapt to future industry demands. Ultimately, implementing this system will enhance efficiency, reduce manual workload, improve customer experience, and support business growth in the competitive airline industry.

6. SCOPE OF THE PROJECT

The airline booking website is an application stored in the user server. The purpose of the website is to resolve the client to allow website users to perform tasks related to visit website, booking tickets and canceling tickets of an airline flight. The system enables to perform the following functions:

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